

Notes: Almeida and Campello (RFS, 2007)
Financial Constraints, Asset Tangibility, and Corporate Investment

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1 Big picture

- **Question.** Do financing frictions affect real corporate investment? The paper asks whether imperfect credit access changes investment behavior, and it proposes a new way to identify that effect.
- **Core idea.** Tangible assets are more pledgeable. If creditors can recover more value in default, the firm can borrow more. More borrowing capacity then allows the firm to invest more in pledgeable assets.
- **Credit multiplier.** A cash-flow shock has a direct effect on investment and an indirect effect through debt capacity. The indirect effect is stronger when assets are more tangible.
- **Key theoretical prediction.** Among financially constrained firms, investment-cash flow sensitivity should increase with asset tangibility. Among unconstrained firms, tangibility should not affect investment-cash flow sensitivity.
- **Nonmonotonicity.** Tangibility both strengthens the multiplier for constrained firms and makes firms more likely to become unconstrained. Therefore the sensitivity is increasing in tangibility only at low or moderate tangibility levels, then disappears once the firm is unconstrained.
- **Empirical setting.** The authors study manufacturing firms in COMPUSTAT from 1985 to 2000. Investment is capital expenditures over beginning-of-period capital stock.
- **Main empirical design.** The investment equation augments the standard investment-cash flow regression with *Tangibility* and $CashFlow \times Tangibility$. The interaction term is the empirical test of the credit multiplier.
- **Constraint classification.** The benchmark approach uses a switching regression with endogenous selection into constrained and unconstrained regimes. The probability of being constrained is estimated jointly with the investment equations.
- **Main result.** Asset tangibility positively affects investment-cash flow sensitivities for constrained firms, but not for unconstrained firms. Tangibility also lowers the probability that a firm is financially constrained.
- **Economic takeaway.** The paper turns the Kaplan-Zingales critique into an identification strategy: sensitivities need not be monotonic in constraints, but their interaction with asset pledgeability reveals financing frictions.

2 Incremental contribution of the paper

- **From sensitivity levels to sensitivity slopes.** Earlier work often compares the level of investment-cash flow sensitivity across constrained and unconstrained firms. This paper focuses on how that sensitivity changes with asset tangibility.
- **A new identification strategy.** The test is difference-in-differences in spirit: compare how tangibility changes the cash-flow sensitivity of constrained firms relative to unconstrained firms.
- **A direct response to Kaplan and Zingales.** Kaplan and Zingales argue that sensitivity is not monotonically related to financing constraints. Almeida and Campello accept the nonmonotonicity, model it, and use it to generate a sharper prediction.

- **A credit-multiplier contribution.** The paper brings the Kiyotaki-Moore collateral multiplier logic into corporate investment regressions. Tangibility is not just a control variable; it is the mechanism through which borrowing capacity expands.
- **A measurement contribution.** The authors construct several tangibility proxies: expected liquidation values of firm assets, industry asset liquidity based on used capital markets, and industry durability based on cyclicalities.
- **An endogenous-constraints contribution.** The benchmark estimator does not rely only on ex ante sample splits. It jointly estimates investment regimes and the probability of financial constraint status.
- **A robustness contribution.** The paper checks whether results survive standard concerns about noisy Q by using Cummins et al., Erickson-Whited, and Bond-Meghir style estimators.
- **A falsification-style contribution.** The paper studies R&D investment. Because R&D does not create tangible collateral in the same direct way, the cash-flow-tangibility interaction should disappear after controlling for fixed investment; this is what the paper finds.
- **A broader corporate-finance contribution.** The findings support the view that financing frictions shape real investment, not only financial policies such as cash holdings or leverage.

3 Data and measurement

3.1 Sample construction

- The sample contains U.S. manufacturing firms with SIC codes 2000–3999.
- The sample period is 1985–2000.
- The data come from COMPUSTAT annual industrial tapes.
- Firms must have data on total assets, market capitalization, capital expenditures, cash flow, and plant, property, and equipment.
- The authors exclude very small firms with capital stock below \$5 million, observations with extreme real asset or sales growth, and observations with negative Q or $Q > 10$.
- Firms must appear for at least three consecutive years, which is necessary for the lag structure and instrumental-variable procedures.

Interpretation: The data restrictions are designed to focus on firms for which standard linear investment equations are meaningful and to avoid major corporate events that mechanically distort investment and size measures.

3.2 Main variables

- **Investment.** Capital expenditures divided by beginning-of-period capital stock.
- Q . Market value of assets divided by book value of assets. This is the main proxy for investment opportunities.

- **CashFlow.** Earnings before extraordinary items plus depreciation, divided by beginning-of-period capital stock.
- **Tangibility.** A proxy for asset pledgeability. Higher tangibility means creditors can recover more value in default.
- **Financial-constraint status.** In the benchmark model, status is estimated endogenously. In complementary tests, status is assigned using payout policy, size, bond ratings, and commercial-paper ratings.

Interpretation: The key variable is not cash flow by itself. The key variable is the interaction between cash flow and tangibility, because that term captures whether pledgeability amplifies investment responses to internal funds.

3.3 Tangibility measures

- **Firm liquidation values.** The baseline measure follows Berger, Ofek, and Swary. It applies recovery weights to receivables, inventories, and fixed capital, adds cash, and scales by total book assets.
- **Industry asset liquidity.** This measure uses Census data on the ratio of used to total fixed depreciable capital expenditures in an industry. A larger used-capital market implies easier redeployment and liquidation.
- **Industry durability.** This measure separates durable and nondurable industries, following Sharpe. Assets in nondurable industries are treated as more liquid because their users are less jointly hit by adverse cyclical shocks.

Interpretation: The use of both firm-level and industry-level measures is important. Firm-level liquidation values are closest to pledgeability, but industry-level measures reduce concerns that a firm's own investment choices mechanically drive measured tangibility.

3.4 Financial constraint measures

- **Switching regression.** The benchmark approach estimates two investment regimes and a selection equation at the same time.
- **Payout policy.** Firms in the bottom three payout deciles are classified as constrained; firms in the top three deciles are classified as unconstrained.
- **Firm size.** Firms in the bottom three asset-size deciles are classified as constrained; firms in the top three deciles are classified as unconstrained.
- **Bond ratings.** Firms with debt but no bond rating are classified as constrained; rated firms are classified as unconstrained.
- **Commercial paper ratings.** Firms with debt but no commercial-paper rating are classified as constrained; firms with commercial-paper ratings are classified as unconstrained.

Interpretation: The paper uses both endogenous and ex ante constraint classifications. This matters because a key part of the theory is that tangibility itself affects whether a firm is constrained.

4 Theoretical model and empirical specifications

4.1 Credit constraint and optimal investment

The firm invests I at date 0 and produces $f(I)$ at date 1. If creditors seize physical assets, they recover only a fraction τ of investment. The borrowing constraint is

$$B \leq \tau I.$$

The firm has internal funds W , so investment satisfies

$$I \leq W + \tau I.$$

The first-best investment level I^{FB} satisfies

$$f'(I^{FB}) = 1.$$

The firm is constrained when

$$\tau < \tau^*(W, I^{FB}) = \max \left\{ 1 - \frac{W}{I^{FB}}, 0 \right\}.$$

Optimal investment is

$$I(W, \tau) = \frac{W}{1 - \tau} \quad \text{if } \tau < \tau^*(W, I^{FB}),$$

and

$$I(W, \tau) = I^{FB} \quad \text{if } \tau \geq \tau^*(W, I^{FB}).$$

Therefore

$$\frac{\partial I}{\partial W} = \frac{1}{1 - \tau} \quad \text{if constrained,}$$

and

$$\frac{\partial I}{\partial W} = 0 \quad \text{if unconstrained.}$$

Interpretation: For constrained firms, a cash-flow shock relaxes the credit constraint directly and indirectly. The indirect part is larger when τ is high, because each extra unit of investment supports more borrowing.

4.2 Economic meaning of nonmonotonicity

- At low levels of tangibility, the firm remains constrained. More tangibility raises the investment response to cash flow because it raises debt capacity.
- At high levels of tangibility, the firm may become unconstrained. Once first-best investment can be financed, investment no longer depends on cash flow.
- Thus, the relationship between tangibility and investment-cash flow sensitivity is positive among constrained firms but disappears among unconstrained firms.

Interpretation: This is the paper's central theoretical insight. Tangibility both amplifies the sensitivity and relaxes the constraint. The empirical design must therefore condition on constraint status.

4.3 Baseline investment equation

The baseline empirical model is

$$Investment_{it} = \alpha_1 Q_{i,t-1} + \alpha_2 CashFlow_{it} + \alpha_3 Tangibility_{it} + \alpha_4 (CashFlow \times Tangibility)_{it} + \sum_i firm_i + \sum_t year_t + \varepsilon_{it}.$$

- α_2 is the cash-flow coefficient when tangibility equals zero.
- α_4 is the key coefficient: it measures whether cash-flow sensitivity increases with tangibility.
- The theory predicts $\alpha_4 > 0$ for constrained firms.
- The theory predicts no positive effect of tangibility on cash-flow sensitivity for unconstrained firms.

Interpretation: Because tangibility is interacted with cash flow, the cash-flow coefficient alone is not the main object. The economically meaningful sensitivity is $\alpha_2 + \alpha_4 Tangibility$.

4.4 Switching regression model

The benchmark estimator assumes two unobserved investment regimes:

$$I_{1it} = X_{it}\alpha_1 + \varepsilon_{1it},$$

$$I_{2it} = X_{it}\alpha_2 + \varepsilon_{2it},$$

$$y_{it}^* = Z_{it}\phi + u_{it}.$$

Observed investment is

$$I_{it} = I_{1it} \quad \text{if } y_{it}^* < 0,$$

and

$$I_{it} = I_{2it} \quad \text{if } y_{it}^* \geq 0.$$

- X_{it} contains Q , cash flow, tangibility, and cash-flow-by-tangibility.
- Z_{it} contains predictors of financial constraint status.
- Z_{it} includes size, age, dividend payout, short-term debt, long-term debt, growth opportunities, bond-rating status, financial slack, and tangibility.
- The model jointly estimates investment behavior and the probability that a firm is in the constrained regime.

Interpretation: The switching regression is attractive because the theory says tangibility itself influences constraint status. A simple ex ante split would ignore that endogenous channel.

4.5 Robustness estimators

- The paper uses analysts' earnings forecasts as instruments for Q following Cummins et al.
- It uses Erickson-Whited style GMM methods that are designed for measurement error in Q .

- It uses Bond-Meghir Euler-equation style specifications that do not rely on Q in the same way.
- The paper also checks an R&D specification. The R&D exercise asks whether a poor proxy for marginal asset pledgeability would falsely produce the same multiplier result.

Interpretation: The robustness tests are aimed at the standard objection that cash flow might proxy for mismeasured investment opportunities rather than financing constraints.

5 Identification and empirical design

5.1 What identifies the credit multiplier

- The identifying variation is not the raw cash-flow sensitivity.
- The identifying variation is whether the cash-flow sensitivity changes with asset tangibility differently across constrained and unconstrained regimes.
- If the result were simply caused by cash flow proxying for investment opportunities, it would be hard to explain why the cash-flow-by-tangibility interaction is positive only among constrained firms.
- If the result were simply caused by high-tangibility firms having different investment opportunities, unconstrained firms should also show similar interaction effects.

Interpretation: The design is a cross-regime comparison of interaction slopes. This is why the paper argues that its approach sidesteps the standard Fazzari et al. sensitivity-level critique.

5.2 Why the Kaplan-Zingales critique does not overturn the test

- Kaplan and Zingales emphasize that investment-cash flow sensitivity need not increase monotonically with the degree of constraint.
- Almeida and Campello's model has exactly that property: once tangibility is high enough, the firm becomes unconstrained and sensitivity falls to zero.
- The test is therefore not that more constrained firms always have higher sensitivity.
- The test is that, conditional on being constrained, pledgeability amplifies the sensitivity.

Interpretation: The paper accepts the nonmonotonicity point but uses a sharper mechanism-based prediction: constrained firms with more pledgeable assets should display larger investment responses to cash-flow shocks.

5.3 Remaining limitations

- The paper is still observational. Cash flow, tangibility, and investment opportunities are not randomly assigned.
- Tangibility of existing assets may not perfectly measure tangibility of marginal new investment.
- Switching-regression classification depends on distributional assumptions.

- The sample is manufacturing firms, so external validity to services, technology firms, or financial firms is not automatic.
- The mechanism is about debt capacity and pledgeability; it may be weaker for firms that rely more on intangible capital or relationship-based financing.

Interpretation: The empirical design is stronger than a simple cash-flow sensitivity comparison, but it does not make the paper a randomized experiment.

6 Tables and figures

The paper contains no standalone graphical figures. The visual material extracted below consists of the paper's tables.

6.1 Table 1 and Table 2: Tangibility measures and sample characteristics

Read:

- Table 1 reports the distribution of the three tangibility proxies.
- The average firm-level liquidation value is 0.526, meaning assets are expected to recover roughly 53 cents per dollar of book value in liquidation.
- The average industry asset liquidity measure is 0.074, meaning used capital acquisitions are a small share of total industry capital acquisitions.
- Industry durability is an indicator, with 46.4% of observations classified as nondurable industries.
- Table 2 compares investment, Q , and cash flow across low- and high-tangibility firms.
- The patterns are not mechanically uniform across measures: high tangibility is associated with higher investment and cash flow under the first two measures, but not under the durability measure.

Economic message: The sample contains meaningful variation in asset pledgeability. The descriptive statistics do not suggest that the main results are simply driven by one obvious tangibility-related difference in investment opportunities.

Table 1
Summary statistics for asset tangibility

	Mean	Median	Std. Dev.	Pct. 10	Pct. 25	Pct. 75	Pct. 90	<i>N</i>
Asset tangibility measures								
1. Firm liquidation values	0.526	0.537	0.169	0.262	0.324	0.686	0.739	17,880
2. Industry asset liquidity	0.074	0.059	0.052	0.026	0.038	0.099	0.136	11,826
3. Industry durability	0.464	0	0.499	0	0	1	1	18,304

This table displays summary statistics for asset tangibility. There are three measures of asset tangibility. The first is based on a firm-level proxy for expected value of assets in liquidation [as in Berger et al. (1996)] and the second on an industry-level measure of asset redeployability based on Census data. These two measures are continuous. The third tangibility measure is based on Sharpe's (1994) industry "durability," where a value of 1 is assigned to firms belonging to nondurable goods industries. The sampled firms include only manufacturers (SICs 2000–3999) and the sample period is 1985 through 2000.

(a) Table 1: Summary statistics for asset tangibility.

Table 2
Summary statistics of investment, Q , and cash flow, across low- and high-tangibility firms

	Investment			Q			CashFlow			<i>N</i>
	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.	
Asset Tangibility Measures										
1. Firm Liquidation Values										
Low-Tangibility Firms	0.219	0.210	0.148	0.997	0.960	0.429	0.319	0.306	0.667	5,364
High-Tangibility Firms	0.266	0.218	0.229	1.024	0.966	0.460	0.401	0.384	0.683	5,364
2. Industry Asset Liquidity										
Low-Tangibility Firms	0.241	0.185	0.208	1.032	0.922	0.371	0.331	0.264	0.547	3,547
High-Tangibility Firms	0.255	0.206	0.198	0.989	0.919	0.400	0.387	0.353	0.495	3,546
3. Industry Durability										
Low-Tangibility Firms	0.252	0.199	0.206	1.008	0.992	0.415	0.342	0.315	0.713	9,782
High-Tangibility Firms	0.216	0.183	0.155	0.988	0.928	0.381	0.323	0.290	0.494	8,522

This table displays summary statistics for investment, Q , and cash flows across groups of low- and high-tangibility firms. *Investment* is defined as the ratio of capital expenditures (COMPUSTAT item #128) to beginning-of-period capital stock (lagged item #8). Q is computed as the market value of assets divided by the book value of assets (= (item #6 + (item #24 × item #25) – item #60 – item #74) / (item #6)). *CashFlow* is earnings before extraordinary items and depreciation (item #18 + item #14) divided by the beginning-of-period capital stock. There are three measures of asset tangibility. The first is based on a firm-level proxy for expected value of assets in liquidation and the second on an industry-level measure of asset redeployment. These two measures are continuous and we define as low-tangibility (high-tangibility) firms those ranked in the bottom (top) three deciles of the tangibility distribution. The third tangibility measure is based on Sharpe's (1994) industry "durability," where low-tangibility (high-tangibility) firms are those in the durables (nondurables) industries. The sampled firms include only manufacturers (SICs 2000–3999) and the sample period is 1985 through 2000.

(b) Table 2: Investment, Q , and cash flow by tangibility.

6.2 Table 3: Switching regressions with endogenous constraint selection

Read:

- Panel A reports the benchmark investment regressions from the switching-regression model.
- For constrained firms, the coefficient on $CashFlow \times Tangibility$ is positive and statistically significant for all three tangibility measures.
- For unconstrained firms, the same interaction is small, insignificant, or negative.
- With the firm liquidation-value measure, the constrained interaction coefficient is 0.4898, while the unconstrained coefficient is -0.0873.
- Panel B reports the selection equation. Tangibility enters negatively and significantly, so more tangible firms are less likely to be assigned to the constrained regime.
- Smaller, younger, low-dividend, unrated, high-growth-opportunity firms are more likely to be classified as constrained.

Economic message: Table 3 is the core evidence. Tangibility amplifies cash-flow sensitivity only where financing frictions should matter, and it simultaneously relaxes the probability of being financially constrained.

Table 3
Investment–Cash Flow Sensitivity and Tangibility: Endogenous Constraint Selection

Dependent Variable	Independent Variables				
Investment	Q	CashFlow	Tangibility	CashFlow× Tangibility	N
Panel A: Main Regressions—Financial constraints assignments use tangibility and the proxies of Hovakimian and Titman (2006)					
Asset Tangibility Measures					
1. <i>Firm Liquidation Values</i>					
Constrained Firms	0.0819** (10.77)	-0.0270** (-2.26)	0.0342 (1.44)	0.4898** (8.99)	17,880
Unconstrained Firms	0.0639** (2.64)	0.0598* (2.27)	0.0176 (0.62)	-0.0873 (-1.46)	
2. <i>Industry Asset Liquidity</i>					
Constrained Firms	0.0801** (7.90)	0.0373** (3.10)	0.0211 (1.26)	0.3923** (5.01)	11,826
Unconstrained Firms	0.0648** (4.06)	0.0905** (4.14)	0.0668 (0.80)	0.0711 (0.82)	
3. <i>Industry Durability</i>					
Constrained Firms	0.0561** (5.44)	0.0849** (3.09)	0.0133 (1.09)	0.0492** (2.88)	18,304
Unconstrained Firms	0.0335** (3.25)	0.1454** (8.65)	0.0197 (1.52)	-0.0147 (-0.86)	

This table displays results from the investment regressions in the switching regression model (Equations (8)–(10) in the text). These equations are estimated with firm- and time-fixed effects. Switching regression estimations allow for endogenous selection into “financially constrained” and “financially unconstrained” categories via maximum likelihood methods. Panel B of Table 3 reports the coefficients from the “regime selection” regressions, where *Tangibility* (various definitions) and the selection variables employed in Hovakimian and Titman (2006) are used to assign firms into constraint categories. All data are from the annual COMPUSTAT industrial tapes. The sampled firms include only manufacturers (SICs 2000–3999) and the sample period is 1985 through 2000. The estimations correct the error structure for heteroskedasticity and clustering using the White-Huber estimator. *t*-statistics (in parentheses). **, * indicate statistical significance at the 1- and 5-percent (two-tail) test levels, respectively.

(a) Table 3, Panel A: Investment equations.

Table 3
(Continued)

Regime Selection Variables	Asset Tangibility Measures		
	Firm Liq. Values	Industry Liquidity	Industry Durability
Panel B: Endogenous Selection Regressions			
<i>LogBookAssets</i>	-0.3248** (-8.97)	-0.2811** (-9.34)	0.2271** (-10.98)
<i>LogAge</i>	-0.4097** (-10.96)	-0.3277** (-15.69)	-0.2355** (-8.24)
<i>DummyDivPayout</i>	-0.1833** (-7.48)	-0.2479** (-11.90)	-0.1976** (-10.50)
<i>ShortTermDebt</i>	0.0919 (0.22)	0.0293 (0.12)	0.0274 (0.05)
<i>LongTermDebt</i>	0.1551 (1.22)	0.0924 (1.02)	0.0714 (0.66)
<i>GrowthOpportunities</i>	0.2609** (9.30)	0.2719** (8.82)	0.2246** (8.01)
<i>DummyBondRating</i>	-0.1757** (-6.02)	-0.2066** (-8.09)	-0.1855** (-5.11)
<i>FinancialSlack</i>	0.2719** (8.42)	0.2980** (9.32)	0.2985** (7.42)
<i>Tangibility</i>	-0.0994** (-7.15)	-0.0766** (-6.98)	-0.1255** (-7.42)
Model <i>P</i> -value (Likelihood Ratio Test)	0.000	0.000	0.000

For each of the estimations reported in this panel (one for each of three proxies for asset tangibility), the dependent variable is coded 1 for assignment into investment regime 1, and 0 for assignment into investment regime 2. As explained in the text, firms assigned into investment regime 1 are classified as financially constrained, and those assigned into investment regime 2 are classified as financially unconstrained. This classification is based on theoretical priors about which firm characteristics are likely to be associated with financial constraints. Each of the selection variables is entered in lagged form. *LogBookAssets* is the natural logarithm of total book assets (item #6). *LogAge* is the natural log of the number of years the firm appears in the COMPUSTAT tapes since 1971. *DummyDivPayout* is a dummy variable that equals 1 if the firm has made any cash dividend payments. *ShortTermDebt* is the ratio of short-term debt (item #34) to total assets. *LongTermDebt* is the ratio of long-term debt (item #9) to total assets. *GrowthOpportunities* is the ratio of market to book value of assets. *DummyBondRating* is a dummy variable that equals 1 if the firm has a bond rating assigned by Standard & Poors. *FinancialSlack* is the ratio of cash and liquid securities (item #1) to lagged assets. *Tangibility* is measured according to various firm- and industry-level measures (see text for details). The estimations correct the error structure for heteroskedasticity and clustering using the White-Huber estimator. *t*-statistics (in parentheses). The last row reports *P*-values for the test of the null hypothesis that a single investment regime is sufficient to describe the data. **, * indicate statistical significance at the 1- and 5-percent (two-tail) test levels, respectively.

(b) Table 3, Panel B: Endogenous selection equations.

6.3 Table 4: Ex ante financial-constraint classifications

Read:

- Table 4 repeats the investment regressions using traditional ex ante constraint classifications.
- Panel A uses firm-level liquidation values. The constrained interaction terms range from 0.3260 to 0.3985 and are significant across all four classification criteria.
- Panel B uses industry asset liquidity. The constrained interaction terms remain positive and significant under payout, size, bond-rating, and commercial-paper classifications.
- Panel C uses industry durability. The constrained interaction terms remain positive and significant or near significant across classifications.
- In contrast, the unconstrained interaction terms are generally insignificant and often negative.

Economic message: The main result does not depend on the switching-regression framework. Even with standard constraint splits, the cash-flow-tangibility interaction appears among constrained firms but not unconstrained firms.

Table 4
Investment–cash flow sensitivity and tangibility: *Ex ante* constraint selection

Dependent Variable	Independent Variables				R^2	N
Investment	Q	CashFlow	Tangibility	CashFlow× Tangibility		
Panel A: Tangibility proxied by firm-level liquidation values (based on Berger et al. (1996))						
Financial Constraints Criteria						
1. <i>Payout Policy</i>						
Constrained Firms	0.0671** (4.35)	−0.0715** (−5.55)	−0.0501 (−0.94)	0.3451** (5.63)	0.118	5,697
Unconstrained Firms	0.0380** (2.72)	0.2042* (2.35)	0.0273 (0.40)	−0.0439 (−0.31)	0.149	5,327
2. <i>Firm Size</i>						
Constrained Firms	0.0623** (3.83)	−0.1066** (−5.46)	−0.1053 (−1.56)	0.3985** (8.14)	0.103	4,854
Unconstrained Firms	0.0543** (3.71)	0.1901* (2.43)	0.0996 (1.32)	−0.1652 (−1.14)	0.114	5,300
3. <i>Bond Ratings</i>						
Constrained Firms	0.0862** (7.64)	−0.0926** (−5.97)	−0.1112 (−1.41)	0.3260** (6.59)	0.106	11,748
Unconstrained Firms	0.0402** (3.73)	0.0623 (0.96)	−0.0177 (−0.35)	0.0953 (0.87)	0.091	5,091
4. <i>Commercial Paper Ratings</i>						
Constrained Firms	0.0841** (8.51)	−0.0897** (−6.03)	−0.1215 (−1.78)	0.3588** (7.06)	0.099	14,194
Unconstrained Firms	0.0353* (2.51)	0.0908* (2.02)	0.0071 (0.12)	0.1177 (1.42)	0.138	2,645

This table displays OLS-FE (firm and year effects) estimation results of the augmented investment model (Equation (7) in the text). The estimations use pre-determined firm selection into “financially constrained” and “financially unconstrained” categories. Constraint category assignments use *ex ante* criteria based on firm dividend payout, size, bond ratings, and commercial paper ratings (see text for details). All data are from the annual COMPUSTAT industrial tapes. The sampled firms include only manufacturers (SICs 2000–3999) and the sample period is 1985 through 2000. The estimations correct the error structure for heteroskedasticity and clustering using the White-Huber estimator. *t*-statistics (in parentheses). **, * indicate statistical significance at the 1- and 5-percent (two-tail) test levels, respectively.

(a) Table 4, Panel A: Firm-level liquidation values.

Table 4
(Continued)

Dependent Variable	Independent Variables				R^2	N
	Investment	Q	CashFlow	Tangibility		
Panel B: Tangibility proxied by industry-level asset liquidity (based on redeployment of used capital)						
Financial Constraints Criteria						
1. <i>Payout Policy</i>						
Constrained Firms	0.0711** (4.05)	0.0591** (2.70)	−0.0359 (−0.41)	0.5781** (3.84)	0.078	3,387
Unconstrained Firms	0.0565** (3.37)	0.0934** (5.58)	0.1276 (1.21)	−0.2328 (−1.15)	0.142	3,117
2. <i>Firm Size</i>						
Constrained Firms	0.0820** (4.30)	0.0881** (3.75)	0.0633 (0.62)	0.2952* (2.28)	0.100	3,104
Unconstrained Firms	0.0753** (4.42)	0.0725** (3.40)	−0.0820 (−0.56)	−0.3448 (−1.08)	0.114	3,117
3. <i>Bond Ratings</i>						
Constrained Firms	0.0914** (7.09)	0.0660* (2.36)	−0.0083 (−0.08)	0.5048** (2.94)	0.117	6,889
Unconstrained Firms	0.0576** (4.03)	0.1078** (3.00)	−0.0799 (−0.75)	0.0607 (0.28)	0.092	3,169
4. <i>Commercial Paper Ratings</i>						
Constrained Firms	0.0900** (7.71)	0.0682** (2.88)	−0.0127 (−0.14)	0.4631** (3.23)	0.109	8,471
Unconstrained Firms	0.0313** (3.22)	0.1552** (6.19)	−0.1371 (−0.85)	0.0454 (0.21)	0.143	1,587

**, * indicate statistical significance at the 1- and 5-percent (two-tail) test levels, respectively.

(b) Table 4, Panel B: Industry asset liquidity.

6.4 Table 4 continued and Table 5: Durability proxy and measurement-error robustness

Read:

- Panel C of Table 4 uses the durable/nondurable industry dichotomy.
- The constrained interaction coefficient is 0.0552 under payout, 0.0668 under size, 0.0491 under bond ratings, and 0.0573 under commercial-paper ratings.
- Table 5 focuses on the coefficient on $CashFlow \times Tangibility$ under estimators designed to address measurement error in Q .

- The Cummins et al. estimator, Erickson-Whited estimator, and Bond-Meghir estimator all preserve the basic pattern: constrained-firm interaction coefficients are positive and significant; unconstrained coefficients are not.

Economic message: The robustness tests speak directly to the concern that cash flow is proxying for unmeasured investment opportunities. The multiplier result survives when the paper uses alternative estimators built for that concern.

Table 4
(Continued)

Dependent Variable	Independent Variables					
Investment	Q	CashFlow	Tangibility	CashFlow× Tangibility	R^2	N
Panel C: Tangibility proxied by product durability (based on Sharpe (1994) industry dichotomy)						
Financial Constraints Criteria						
1. <i>Payout Policy</i>						
Constrained Firms	0.0738** (7.53)	0.0501** (3.62)	−0.0290 (−1.73)	0.0552** (2.97)	0.095	6,443
Unconstrained Firms	0.0220 (1.59)	0.1602** (8.80)	0.0050 (0.52)	−0.0388 (−1.54)	0.233	5,497
2. <i>Firm Size</i>						
Constrained Firms	0.0537** (4.15)	0.1208** (5.80)	−0.0150 (−1.45)	0.0668** (3.27)	0.099	5,497
Unconstrained Firms	0.0546** (4.28)	0.0740** (3.40)	−0.0111 (−0.90)	−0.0575* (−2.37)	0.249	5,497
3. <i>Bond Ratings</i>						
Constrained Firms	0.1051** (8.51)	0.0737** (4.91)	−0.0403* (−2.43)	0.0491* (2.14)	0.088	13,213
Unconstrained Firms	0.0314** (3.12)	0.1326** (6.10)	−0.0127 (−1.40)	−0.0281 (−0.95)	0.185	5,091
4. <i>Commercial Paper Ratings</i>						
Constrained Firms	0.0958** (8.58)	0.0851** (6.49)	−0.0446** (−2.36)	0.0573** (3.03)	0.095	15,659
Unconstrained Firms	0.0151 (1.35)	0.1333** (6.01)	−0.0020 (−0.14)	−0.0394* (−2.20)	0.226	2,645

**, * indicate statistical significance at the 1- and 5-percent (two-tail) test levels, respectively. Regressions in this panel include only year effects.

(a) Table 4, Panel C: Industry durability.

Table 5
Robustness Checks: Measurement Errors in Q

Dependent Variable	Financial Constraints Criteria			
Investment	Payout Policy	Firm Size	Bond Ratings	CP Ratings
1. <i>Cummins et al. Estimator</i>				
Constrained Firms	0.2976** (3.19)	0.4420** (5.78)	0.2474** (2.99)	0.2174** (2.63)
Unconstrained Firms	−0.0752 (−0.46)	−0.2413 (−1.51)	−0.0470 (−0.36)	0.0035 (0.67)
2. <i>Erickson-Whited Estimator</i>				
Constrained Firms	N/A	0.4710** (2.91)	N/A	0.2490** (2.66)
Unconstrained Firms	N/A	−0.1230 (−0.51)	N/A	−0.0240 (−0.08)
3. <i>Bond-Meghir Estimator</i>				
Constrained Firms	0.3123** (2.83)	0.5109** (3.69)	0.3566** (3.77)	0.3243** (4.10)
Unconstrained Firms	0.0111 (0.43)	0.0164 (0.45)	0.0554 (0.86)	−0.0477 (−0.76)

This table reports results from estimators used to address measurement errors in the proxy for investment opportunities, Q , from the baseline regression model (Equation 7) in the text. Each cell displays the estimates of the coefficients returned for *CashFlow*×*Tangibility* and the associated test statistics. The estimations use pre-determined firm selection into “financially constrained” and “financially unconstrained” categories. *Tangibility* is the firm-level proxy for asset liquidation values (based on Berger et al. (1996)). The first set of estimations (row 1) follows Cummins et al. (1999) GMM procedure where Q is instrumented with analysts’ earnings forecasts. The second set of estimations (row 2) uses the Erickson–Whited (2000) GMM5 estimator. The estimations of row 3 are based on Bond and Meghir (1994), where lags of investment, sales, and debt are added as controls and instrumented (estimated via GMM). All estimations control for firm- and year-fixed effects. All data are from the annual COMPUSTAT industrial tapes. The sampled firms include only manufacturers (SICs 2000–3999) and the sample period is 1985 through 2000. The estimations correct the error structure for heteroskedasticity and clustering. *t*-statistics (in parentheses). Hansen’s *J*-statistics for overidentifying restrictions tests [in square brackets]. **, * indicate statistical significance at the 1- and 5-percent (two-tail) test levels, respectively.

(b) Table 5: Robustness checks for measurement errors in Q .

6.5 Table 6: Fixed capital and R&D expenditures

Read:

- Table 6 estimates an R&D expenditure model in which fixed capital investment is instrumented.
- R&D is strongly associated with predicted fixed investment.
- Once fixed investment is controlled for, the *CashFlow* × *Tangibility* coefficients are negative and mostly insignificant.
- This is exactly what the authors expect if the collateral multiplier works through tangible fixed assets rather than through R&D.

Economic message: Table 6 functions like a falsification test. If tangibility were merely picking up generic investment opportunities, it could also amplify R&D sensitivity. Instead, the multiplier appears where tangible collateral is created.

Table 6
Fixed capital and R&D expenditures

Dependent Variable	Independent Variables				
R&D Expenditures	$\hat{I}$	CashFlow	Tangibility	CashFlow $\times$ Tangibility	N
Main Regressions—Financial constraints assignments use tangibility and the proxies of Hovakimian and Titman (2006)					
Asset Tangibility Measures					
1. <i>Firm Liquidity Values</i>					
Constrained Firms	0.5298** (2.58)	0.0605* (2.25)	−0.0019 (−0.60)	−0.0816 (−1.36)	15,077
Unconstrained Firms	0.3848* (2.09)	0.1699** (4.27)	−0.0673 (−1.52)	−0.0623 (−1.10)	
2. <i>Industry Asset Liquidity</i>					
Constrained Firms	0.6673** (3.35)	0.0557* (2.00)	0.0296 (1.29)	−0.0969 (−1.29)	10,678
Unconstrained Firms	0.3249* (2.08)	0.1300** (3.27)	0.0275 (1.20)	−0.1169 (−1.45)	
3. <i>Industry Durability</i>					
Constrained Firms	0.6897** (3.77)	0.0444 (1.75)	−0.0094 (−0.66)	−0.0265 (−1.83)	16,527
Unconstrained Firms	0.5614** (2.66)	0.0697* (2.22)	0.0257 (1.14)	−0.0078 (−0.61)	

This table displays results from the R&D expenditures model (Equation (12)), where *Investment* is instrumented ($\hat{I}$). Switching regression estimations allow for endogenous selection into “financially constrained” and “financially unconstrained” categories via maximum likelihood methods. The table reports only the coefficients from the “main” investment regressions. The “regime selection” regressions use *Tangibility* (various definitions) and the selection variables employed in Hovakimian and Titman (2006) to assign firms into constraint categories. All data are from the annual COMPUSTAT industrial tapes. The sampled firms include only manufacturers (SICs 2000–3999) and the sample period is 1985 through 2000. The estimations correct the error structure for heteroskedasticity and clustering using the White-Huber estimator. *t*-statistics (in parentheses). **, * indicate statistical significance at the 1- and 5-percent (two-tail) test levels, respectively.

Table 6: Fixed capital and R&D expenditures.

7 Short summary

- The paper studies whether financing frictions affect real investment.
- The theoretical mechanism is a collateral or credit multiplier: pledgeable assets support borrowing, which supports more investment.
- The model predicts that investment-cash flow sensitivity increases with asset tangibility only for constrained firms.
- Tangibility also reduces the probability that a firm is constrained, creating a nonmonotonic relation between tangibility and sensitivity.
- The empirical setting is U.S. manufacturing firms from 1985 to 2000.
- The key regression term is $CashFlow \times Tangibility$.
- The benchmark switching regression finds positive and significant interaction effects for constrained firms, but not unconstrained firms.
- Tangibility enters negatively in the selection equation, meaning more tangible firms are less likely to be financially constrained.
- Traditional constraint splits based on payout, size, bond ratings, and commercial-paper ratings produce similar results.
- Robustness checks for measurement error in Q preserve the main pattern.

- R&D tests show that the effect does not appear when the proxy should be a poor measure of marginal pledgeability.
- The paper’s main contribution is to identify financing frictions through the interaction between internal funds and asset pledgeability, rather than through sensitivity levels alone.

8 Conclusion

8.1 Core conclusion

Almeida and Campello show that financing frictions affect corporate investment through asset pledgeability. When firms are financially constrained, tangible assets strengthen the response of investment to cash flow because those assets increase borrowing capacity. When firms are unconstrained, tangibility does not meaningfully change investment-cash flow sensitivity.

8.2 Economic intuition

The economic intuition is a multiplier. A positive cash-flow shock allows a constrained firm to invest more. If the investment creates pledgeable assets, the firm can also borrow more against the new assets. The firm can then invest still more. This amplification is stronger when assets are more tangible because creditors recover more value in default.

8.3 Incremental contribution revisited

The paper changes how investment-cash flow sensitivities should be interpreted. It does not claim that higher sensitivities always mean tighter constraints. Instead, it says that sensitivities can reveal constraints when they move in the theoretically predicted way with an external-finance-capacity shifter such as tangibility. This is a major contribution because it answers the critique that sensitivity levels alone are not reliable indicators of financial constraints.

8.4 Implications for corporate finance

The findings imply that asset composition matters for investment policy. Firms with more pledgeable assets can transform cash-flow shocks into larger investment changes because their debt capacity is more procyclical. This helps explain why collateral values and asset markets can amplify corporate investment cycles.

8.5 Implications for empirical work

The paper suggests that researchers should look for mechanism-based interactions rather than relying only on sample splits. A financial-constraint proxy is more persuasive when it changes investment behavior in a direction predicted by a theory of financing frictions.

8.6 Limitations

The paper remains an observational study. Tangibility may correlate with unobserved production technology, investment opportunities, or industry characteristics. The authors address these concerns with alternative tangibility proxies, switching regressions, traditional splits, and measurement-error-robust estimators, but no observational design can eliminate all concerns. The sample is also limited to manufacturing firms, where tangible capital is especially important.

8.7 Takeaway

The enduring takeaway is that pledgeability is an economically meaningful channel through which financing frictions shape real investment. Constrained firms with tangible assets do not simply invest more because they have more cash flow; they invest more because cash flow relaxes credit constraints and expands borrowing capacity.